THE RELATIVE DUALIZING COMPLEX OF A PROPER FLAT MORPHISM IS A RELATIVE DUALIZING COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E2Z (sources/stacks-duality/pinned-original-source.tex line 7530).

Allowed standard prerequisites (editorial).

- The relative dualizing complex is S-perfect. Same chapter, Lemma labelled lemma-properties-relative-dualizing, cited at proof lines 7540-7542. Gives the S-perfectness required of a relative dualizing complex.
- The right adjoint of pushforward for the diagonal. Same chapter, Lemma labelled lemma-twisted-inverse-image, cited at proof lines 7542-7546. Supplies the functor c applied to the pulled back dualizing complex.
- The right adjoint of a closed immersion and the exact support sheaf. Same chapter, Lemmas labelled lemma-twisted-inverse-image-closed and lemma-sheaf-with-exact-support-ext, cited at proof lines 7552-7556. Identify Delta_*(c(...)) with RSheafHom against Delta_*O_X.

Source reference. Original reference 0B6S: sources/stacks-duality/pinned-original-source.tex, source lines 2945–2945. Same-chapter pointer into the author environment remark at source line 2945; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0E2Q: sources/stacks-duality/pinned-original-source.tex, source lines 3166–3166. Same-chapter pointer into the author environment equation at source line 3166; invoked at proof line 7546 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 0E2T: sources/stacks-duality/pinned-original-source.tex, source lines 7234–7234. Same-chapter pointer into the author environment definition at source line 7234; invoked at proof line None of the counted

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proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E4L: sources/stacks-duality/pinned-original-source.tex, source lines 3028–3028. Same-chapter pointer into the author environment lemma at source line 3028; invoked at proof line 7541 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 7544 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9X: sources/stacks-duality/pinned-original-source.tex, source lines 2467–2467. Same-chapter pointer into the author environment lemma at source line 2467; invoked at proof line 7555 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A77: sources/stacks-duality/pinned-original-source.tex, source lines 2339–2339. Same-chapter pointer into the author environment lemma at source line 2339; invoked at proof line 7556 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \to S$ be a proper, flat morphism of finite presentation. The relative dualizing complex $\omega^\bullet_{X/S}$ of Remark 0B6S together with (0E2Q) is a relative dualizing complex in the sense of Definition 0E2T.*

Proof. In Lemma 0E4L we proved that $\omega^\bullet_{X/S}$ is S-perfect. Let c be the right adjoint of Lemma 0A9E for the diagonal $\Delta : X \to X \times_S X$. Then we can apply Δ_* to (0E2Q) to get an isomorphism

$$\Delta_*\mathcal{O}_X \to \Delta_*(c(L\mathrm{pr}_1^*\omega^\bullet_{X/S})) = R\mathcal{H}om_{\mathcal{O}_{X\times_S X}}(\Delta_*\mathcal{O}_X, L\mathrm{pr}_1^*\omega^\bullet_{X/S})$$

The equality holds by Lemmas 0A9X and 0A77. This finishes the proof. □